ARGUS — Master AI Proctoring Architecture & Implementation Specification

Authoritative, unified master specification for the production-ready hybrid CV+VLM surveillance pipeline. This document combines **sub-second situational awareness** (YOLOv8 + MediaPipe), **semantic detection & conclusive multi-frame video verification** (Qwen3-VL), **engineered hardware mitigations**, and **reinforcement learning / continuous adaptation from proctor feedback**.

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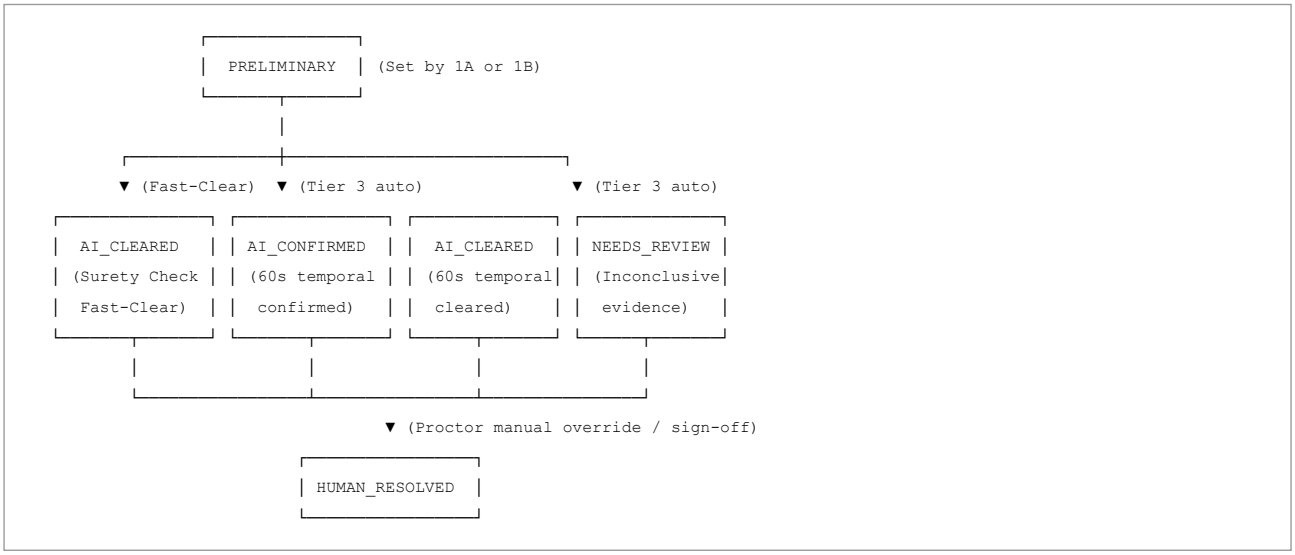
1. Core Pipeline Architecture & Escalation Model

Tier 1 is a **hybrid layer with three sub-components**: a continuous CV engine, an independent adaptive VLM sweep, and a fast async "surety check" that filters CV's known false-positive patterns before they consume proctor attention or Tier 3 compute.


[LIVE RTSP/CCTV FEED — per desk]
TIER 1A: CV Detection
YOLOv8 + MediaPipe
every frame, ~20-40ms
geometric triggers:
object shapes, hand-
to-ear distance, head-
turn angle, hand-lap
motion. High recall.
TIER 1B: VLM Frame Sweep
Adaptive Round-Robin
(~1 check per 10-15s/desk)
Catches note CONTENT,
ambiguous context,
communication semantics.
Already carries its own
semantic verdict.
raw candidate
already-verified flag
INSTANT VLM SURETY CHECK
1 frame, 4-field schema
~1.0-1.5s resolution
(Evaluates before alert)
Anomaly created in active feed
status = PRELIMINARY
(shows "Verifying with AI...")
Surety Check Result (1A flags):
High-confidence False Positive (>=0.92)
status: AI_CLEARED
(Routes directly to Cleared Logs,
zero proctor flicker, skips Tier 3)
Plausible / Uncertain (stays in feed)
(Tier 3 fires at +35s)
Wait until 25s-before / 35s-after source footage exists in
the 75s JPEG rolling ring buffer
TIER 3: Auto-Enrichment Video Pass
Extracts 20-24 frames over ≥60s, builds a chronological
narrative reconstruction (BEFORE / DURING / AFTER)
AI_CONFIRMED
Violation
confirmed;
high-res proof
pinned
AI_CLEARED
Benign motion
auto-cleared;
kept in audit
log, filtered
from feed
NEEDS_REVIEW
Ambiguous —
routed to
proctor (3b
Deep Analysis
available)

2. Anomaly Status Lifecycle & State Machine

Every anomaly card maintains a single immutable `event_id`. Updates mutate status and append records in place — no card duplication, no UI jumpiness.



Status	Badge Color	Description	UI Action
`PRELIMINARY`	Amber / Pulsing	Plausible candidate; 60s verification underway.	`"Verifying with AI..."` spinner badge.
`AI_CONFIRMED`	Red / Crimson	Tier 3's 60s temporal pass confirmed illicit behavior.	Pinned high-res proof screenshot; prominent alert.
`AI_CLEARED`	Emerald / Green	Either the async surety check fast-cleared an obvious false alarm, or Tier 3 proved benign context.	Auto-filtered from active feed; retained in audit log — never hard-deleted.
`NEEDS_REVIEW`	Purple / Violet	Tier 3 returned `INCONCLUSIVE`.	Surfaced to proctor; `"Deep Analysis"` (3b) button active.
`HUMAN_RESOLVED`	Slate / Blue	Final decision signed off by an invigilator.	Locked, with proctor signature + review notes.

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3. Standardized Production Prompts Specification

Core architectural rules applied to every prompt below:

1. **`reasoning` / `quick_reasoning` / `independent_observation` is the first key** in every JSON schema, so the model reasons before committing to booleans/verdicts.
2. **`original_confidence` is withheld** from re-analysis prompts (Tier 2, Tier 3) to eliminate anchoring bias.
3. **Accessibility carve-outs** (hearing aids, headwear, occlusion) are restated in every prompt independently.

Tier 1A Support — Instant Surety Check Prompt

Fires automatically on Tier 1A (CV) candidates.

```
def build_instant_surety_prompt(candidate_category: str, cv_confidence: float, bbox: dict | None = None) -> str:
    bbox_line = ""
    if bbox:
        bbox_line = (
            f"The CV detector flagged a region roughly at normalized coordinates "
            f"x={bbox['x']:.2f}, y={bbox['y']:.2f}, w={bbox['w']:.2f}, h={bbox['h']:.2f} — "
            "pay closest attention there, but consider the full frame.\n"
        )
    return (
        "You are a fast automatic verification pass on a computer-vision (YOLOv8/MediaPipe) detection "
        "from a live exam surveillance feed. The CV detector is fast but geometry-based and prone to "
        "specific false positives (e.g. mistaking a pen/ruler for a phone, or ear-touching for an earbud). "
        "Your job is a QUICK confirm/deny check, not a full forensic review.\n"
        "\n"
        f"CV candidate category: {candidate_category}\n"
        f"CV confidence: {cv_confidence:.2f}\n"
        f"{bbox_line}"
        "\n"
        "Quick rule check:\n"
        "- Pens/pencils/erasers/rulers/calculators are NOT phones.\n"
        "- Touching/scratching the ear, adjusting glasses/mask/headwear, or hearing aids/cochlear "
        "implants are NOT earbuds.\n"
        "- Looking at own paper, writing, or brief incidental glances are NOT peeking.\n"
        "- If genuinely unclear from this single frame, say so — do not guess.\n"
        "\n"
        "Return ONLY a raw JSON object:\n"
        "{\n"
        '  "quick_reasoning": "1 short sentence",\n'
        '  "plausible": boolean,\n'
        '  "confidence": float between 0.0 and 1.0,\n'
        '  "visibility": "clear" | "partial" | "obstructed"\n'
        "}\n"
        "\n"
        "plausible = true means the CV flag is worth surfacing for the full pipeline (stays PRELIMINARY, "
        "Tier 3 still runs at +35s). plausible = false with confidence >= 0.92 means this looks like a "
        "clear CV false positive — log it as AI_CLEARED, do not keep it as an active alert."
    )
```

Tier 1B — Independent Adaptive VLM Sweep Prompt (Scoped)

Runs on an adaptive round-robin schedule (~1 check per 10–15s per desk).

```
LIVE_ANALYSIS_PROMPT_SCOPED_NOTES_AND_COMMUNICATION = (
    "You are an AI exam proctor analyzing a single surveillance snapshot of a student at an exam desk. "
    "A separate computer-vision system already handles phone and earbud detection — your job is ONLY "
    "the categories CV geometry cannot reliably judge:\n"
    "1. UNAUTHORIZED_NOTES: Loose cheat sheets, unauthorized paper slips, or notes concealed under desk/lap "
    "— distinguish from the candidate's own exam paper or authorized reference sheet.\n"
    "2. COMMUNICATION: Actively turning and speaking or gesturing to another person — distinguish from "
    "talking to self, mouthing words while reading, or incidental posture shifts.\n"
    "\n"
    "CRITICAL RULES — NORMAL, NOT cheating: writing/reading own exam paper; brief incidental glances; "
    "talking to self while reading.\n"
    "\n"
    "Return ONLY a raw JSON object:\n"
    "{\n"
    '  "reasoning": "1-2 sentences justifying each flag below",\n'
    '  "visibility": "clear" | "partial" | "obstructed",\n'
    '  "notes_detected": boolean,\n'
    '  "notes_confidence": float between 0.0 and 1.0,\n'
    '  "talking_detected": boolean,\n'
    '  "talking_confidence": float between 0.0 and 1.0,\n'
    '  "description": "1 brief sentence summarizing visible student activity",\n'
    '  "confidence": float between 0.0 and 1.0\n'
    "}"
)
```

Tier 2 — Single-Frame Forensic Re-Analysis Prompt (Manual)

```
FORENSIC_REANALYSIS_PROMPT = (
    "You are an independent AI forensic reviewer conducting a SECOND-OPINION verification of a previously "
    "flagged exam anomaly. Your job is to look at this image freshly and form your own conclusion — you "
    "are checking the original detection's work, not confirming it.\n"
    "\n"
    "STEP 1 — Independent observation (do this BEFORE looking at the prior flag below):\n"
    "Describe exactly what you see regarding the candidate's hands, desk surface, lap, ears, and gaze "
    "direction. Note object shapes, positions, and any ambiguity from angle, lighting, or occlusion.\n"
    "\n"
    "STEP 2 — Prior detection to verify (reference only, not ground truth):\n"
    "- Flagged category: {event_type}\n"
    "- Original system description: {description}\n"
    "Only after forming your own observation in Step 1, compare it against this claim. The original "
    "detection may be correct, mistaken, or mislabeled — you have no obligation to agree with it.\n"
    "\n"
    "STEP 3 — Apply these standing rules (NORMAL behavior, NOT violations):\n"
    "- Holding a pen, pencil, eraser, ruler, or calculator is NOT a phone.\n"
    "- Looking down at the candidate's own exam paper is NOT peeking.\n"
    "- Touching hair/face, adjusting glasses/mask/hijab/turban, or resting chin on hand is NOT a hidden earbud.\n"
    "- Hearing aids and cochlear implants are NEVER a WEARABLE_AUDIO violation.\n"
    "- If ears, hands, or desk are not clearly visible due to angle, headwear, or obstruction, this is a "
    "visibility limitation, not evidence of concealment.\n"
    "- Talking to self or mouthing words while reading is NOT communication with another person.\n"
    "\n"
    "STEP 4 — Deliver a verdict:\n"
    "- CONFIRMED_VIOLATION: Clear, unambiguous visual evidence supports the flagged category.\n"
    "- FALSE_POSITIVE: The flagged category does not match what is actually visible; behavior is benign.\n"
    "- CATEGORY_MISMATCH: A violation may be occurring, but it does not match the originally flagged "
    "category (e.g. flagged as PHONE_DETECTED but you observe UNAUTHORIZED_NOTES instead).\n"
    "- INCONCLUSIVE: Image quality, angle, or occlusion genuinely prevents a determination either way.\n"
    "\n"
    "Return ONLY a raw JSON object matching this schema, in this field order:\n"
    "{\n"
    '  "independent_observation": "1-3 sentences describing what you actually see, written before considering the prior flag",\n'
    '  "visibility": "clear" | "partial" | "obstructed",\n'
    '  "inconclusive_reason": "string or null — required if verdict is INCONCLUSIVE",\n'
    '  "verdict": "CONFIRMED_VIOLATION" | "FALSE_POSITIVE" | "CATEGORY_MISMATCH" | "INCONCLUSIVE",\n'
    '  "actual_category": "string or null — only set if verdict is CATEGORY_MISMATCH",\n'
    '  "confidence": float between 0.0 and 1.0,\n'
    '  "forensic_reasoning": "detailed breakdown connecting Step 1 observation to final verdict",\n'
    '  "detected_objects": ["pen", "paper", ...]\n'
    "}\n"
    "\n"
    "confidence must reflect your actual certainty in the verdict. If uncertain, choose INCONCLUSIVE rather than guessing."
)
```

Tier 3 & 3b — Multi-Frame Temporal Window Prompt (Auto & Deep)

```
def build_multiframe_prompt(frame_timestamps: list[float], event_type: str, description: str) -> str:
    frame_index_map = "\n".join(
        f"  Frame {i}: {t:.1f}s" for i, t in enumerate(frame_timestamps)
    )

    return (
        "You are an independent AI forensic reviewer analyzing a chronological sequence of frames "
        f"sampled from a {frame_timestamps[-1]:.0f}-second video clip surrounding a flagged exam anomaly. "
        "Your task is to judge the FULL narrative — before, during, and after — not a single instant.\n"
        "\n"
        "SAMPLED FRAMES (use ONLY these indices/timestamps when referencing time — never invent a "
        "timestamp that isn't listed below):\n"
        f"{frame_index_map}\n"
        "\n"
        "PRIOR DETECTION (reference only, not ground truth):\n"
        f"- Flagged category: {event_type}\n"
        f"- Original single-frame description: {description}\n"
        "Form your own reading of the sequence before weighing this claim — it may be correct, "
        "mistaken, or capturing only a misleading mid-action moment.\n"
        "\n"
        "STEP 1 — Narrative reconstruction:\n"
        "Describe what happens across three phases: BEFORE (leading up to the flagged moment), "
        "DURING (the flagged moment itself), and AFTER (what follows). Note any motion, object "
        "handling, posture changes, or interactions with others across the sequence.\n"
        "\n"
        "STEP 2 — Standing rules (NORMAL behavior, NOT violations):\n"
        "- Stretching, adjusting posture, or reaching for a dropped pen/eraser.\n"
        "- Holding a pen, pencil, eraser, ruler, or calculator — not a phone.\n"
        "- Looking down at, or turning pages of, the candidate's OWN exam paper.\n"
        "- Adjusting glasses, mask, hijab, turban, or hair; scratching the ear.\n"
        "- Hearing aids or cochlear implants — never a WEARABLE_AUDIO violation.\n"
        "- Brief, incidental glances that don't fixate on a neighbor's desk/paper.\n"
        "- If any phase is unclear due to angle, motion blur, or occlusion, say so explicitly rather "
        "than inferring intent.\n"
        "\n"
        "STEP 3 — Verdict:\n"
        "- CONFIRMED_VIOLATION: The sequence shows clear, sustained evidence of the flagged behavior.\n"
        "- FALSE_POSITIVE: The full sequence shows benign behavior; the original flag misread a "
        "single ambiguous moment.\n"
        "- CATEGORY_MISMATCH: A violation is visible, but it doesn't match the originally flagged "
        "category.\n"
        "- INCONCLUSIVE: Even across the full sequence, evidence is genuinely insufficient.\n"
        "\n"
        "STEP 4 — Proof frame selection:\n"
        "Select the frame index (from the list above) that MOST CLEARLY shows the violation, and the "
        "index range that captures its full extent. Do not select a frame or timestamp not listed above.\n"
        "\n"
        "Return ONLY a raw JSON object matching this schema, in this field order:\n"
        "{\n"
        '  "narrative": {\n'
        '    "before": "1-2 sentences",\n'
        '    "during": "1-2 sentences",\n'
        '    "after": "1-2 sentences"\n'
        '  },\n'
        '  "motion_summary": "1 sentence noting any abrupt or notable transitions between frames",\n'
        '  "verdict": "CONFIRMED_VIOLATION" | "FALSE_POSITIVE" | "CATEGORY_MISMATCH" | "INCONCLUSIVE",\n'
        '  "actual_category": "string or null — only if verdict is CATEGORY_MISMATCH",\n'
        '  "inconclusive_reason": "string or null — required if verdict is INCONCLUSIVE",\n'
        '  "proof_frame_index": integer or null,\n'
        '  "proof_range_start_index": integer or null,\n'
        '  "proof_range_end_index": integer or null,\n'
        '  "confidence": float between 0.0 and 1.0,\n'
        '  "forensic_reasoning": "how the narrative supports the verdict",\n'
        '  "detected_objects": ["pen", "paper", ...]\n'
```

```
        "}\n"
        "\n"
        "confidence must reflect genuine certainty across the full sequence, not just the single most "
        "suspicious-looking frame."
    )
```

Interactive Custom Proctor Query Prompt

```
CUSTOM_QUERY_PROMPT_TEMPLATE = (
    "You are an independent AI forensic reviewer. An invigilator has asked a specific question about "
    "this exam surveillance snapshot:\n"
    "\n"
    "Question: {custom_question}\n"
    "\n"
    "Answer only based on what is visibly verifiable in the image. Apply the same standing rules as normal "
    "exam-hall assessment: writing/reading own paper is normal; pens/rulers/calculators are not phones; "
    "hearing aids are not earbuds; occluded views should be reported as such, not guessed at.\n"
    "\n"
    "Return ONLY a raw JSON object:\n"
    "{\n"
    '  "answer": "direct, concise answer to the question",\n'
    '  "confidence": float between 0.0 and 1.0,\n'
    '  "supporting_observation": "what in the image supports this answer",\n'
    '  "visibility": "clear" | "partial" | "obstructed"\n'
    "}"
)
```

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4. Backend Engineering & Concrete Implementation

Memory-Efficient Rolling Video Buffer (JPEG Compression)

> *[!IMPORTANT]* ***Engineering Mitigation (RAM Explosion)****: Storing raw uncompressed NumPy arrays for 75 seconds at 30 FPS consumes ~14 GB of RAM per camera. This implementation encodes frames to JPEG byte strings before buffering, reducing the footprint to* ***~350 MB per camera*** *(a 97.5% memory reduction).*

```
class RollingVideoBuffer:
    """Retains 75 seconds of rolling video as compressed JPEG bytes (350MB RAM footprint)."""
    def __init__(self, fps: int = 30, history_seconds: int = 75, jpeg_quality: int = 80):
        self.fps = fps
        self.capacity = fps * history_seconds  # 2,250 frames
        self.jpeg_quality = jpeg_quality
        self.buffers: Dict[int, deque] = {}      # camera_id -> deque of bytes
        self.timestamps: Dict[int, deque] = {}   # camera_id -> deque of float timestamps
        self._lock = threading.Lock()

    def add_frame(self, camera_id: int, frame: np.ndarray, timestamp: float):
        # Encode to JPEG bytes before appending to buffer
        success, buf = cv2.imencode('.jpg', frame, [cv2.IMWRITE_JPEG_QUALITY, self.jpeg_quality])
        if not success:
            return
        jpeg_bytes = buf.tobytes()

        with self._lock:
            if camera_id not in self.buffers:
                self.buffers[camera_id] = deque(maxlen=self.capacity)
                self.timestamps[camera_id] = deque(maxlen=self.capacity)
            self.buffers[camera_id].append(jpeg_bytes)
            self.timestamps[camera_id].append(timestamp)

    def extract_window(self, camera_id: int, start_time: float, end_time: float) -> Tuple[List[bytes], List[float]]:
        with self._lock:
            if camera_id not in self.buffers:
                return [], []
            frames_bytes, times = [], []
            for b, t in zip(self.buffers[camera_id], self.timestamps[camera_id]):
                if start_time <= t <= end_time:
                    frames_bytes.append(b)
                    times.append(t)
            return frames_bytes, times
```

Anti-Hallucination Timestamp Indexing & Raw Proof Extraction

```
def extract_temporal_proof(
    clip_path: str,
    sampled_times: List[float],
    proof_index: int,
    exam_id: str,
    track_id: int
) -> Optional[str]:
    """Slices the raw, highest-resolution uncompressed frame from disk at the exact AI index."""
    if proof_index is None or proof_index >= len(sampled_times):
        return None

    exact_sec = sampled_times[proof_index]
    cap = cv2.VideoCapture(clip_path)
    cap.set(cv2.CAP_PROP_POS_MSEC, exact_sec * 1000)
    ret, frame = cap.read()
    cap.release()

    if not ret:
        return None

    out_dir = Path(settings.SCREENSHOTS_STORAGE_PATH) / f"exam_{exam_id}"
    out_dir.mkdir(parents=True, exist_ok=True)
    out_path = out_dir / f"PROOF_t{exact_sec:.1f}s_trk{track_id}.jpg"
    cv2.imwrite(str(out_path), frame, [cv2.IMWRITE_JPEG_QUALITY, 92])
    return str(out_path)
```

FastAPI Endpoints with Rate Limiting & Error Handling

```
class AnomalyStatus(str, Enum):
    PRELIMINARY = "PRELIMINARY"
    AI_CONFIRMED = "AI_CONFIRMED"
    AI_CLEARED = "AI_CLEARED"
    NEEDS_REVIEW = "NEEDS_REVIEW"
    HUMAN_RESOLVED = "HUMAN_RESOLVED"

class AnomalySource(str, Enum):
    CV_1A = "CV_1A"
    VLM_SWEEP_1B = "VLM_SWEEP_1B"

MAX_TIER2_RERUNS = 3
MAX_TIER3B_RERUNS = 1

@app.post("/api/events/{event_id}/rerun-analysis")
async def manual_tier2_rerun(
    event_id: str,
    req: ReanalysisRequest = Body(default_factory=ReanalysisRequest),
    current_user: User = Depends(require_any_authenticated),
    db: Session = Depends(get_db)
):
    event = get_event(db, event_id)
    if not event:
        raise HTTPException(status_code=404, detail="Event not found")

    # Concurrency lock
    if getattr(event, "is_reanalyzing", False):
        raise HTTPException(status_code=409, detail="Analysis already in progress for this event")

    # Server-side rate limit enforcement
    history = json.loads(event.analysis_data).get("history", []) if event.analysis_data else []
    tier2_count = sum(1 for h in history if h.get("tier") == "TIER_2_MANUAL")
    if tier2_count >= MAX_TIER2_RERUNS:
        raise HTTPException(status_code=429, detail=f"Tier 2 re-run limit ({MAX_TIER2_RERUNS}) reached for this event")

    event.is_reanalyzing = True
    db.commit()

    try:
        result = await vlm.execute_tier2_verification(
            image_path=event.evidence_screenshot_path,
            event_type=event.event_type.value,
            description=event.description,
            custom_prompt=req.custom_prompt
        )

        verification_record = {
            "tier": "TIER_2_MANUAL",
            "triggered_by": current_user.username,
            "timestamp": datetime.now(timezone.utc).isoformat(),
            "result": result
        }
        history.append(verification_record)
        event.analysis_data = json.dumps({
            "latest_verification": result,
            "history": history
        })

        db.commit()
        await broadcast_event(str(event.exam_id), event)
        return _event_to_response(event)

    except Exception as exc:
        error_record = {
```

```
            "tier": "TIER_2_MANUAL",
            "triggered_by": current_user.username,
            "timestamp": datetime.now(timezone.utc).isoformat(),
            "error": str(exc)
        }
        history.append(error_record)
        event.analysis_data = json.dumps({
            "latest_verification": {"status": "ERROR", "detail": str(exc)},
            "history": history
        })
        db.commit()
        await broadcast_event(str(event.exam_id), event)
        raise HTTPException(status_code=502, detail="VLM analysis failed — retry available") from exc

    finally:
        event.is_reanalyzing = False
        db.commit()
```

Surety-Check Dispatcher & Fast-Clear Guardrails

```
NON_DISMISSIBLE_CATEGORIES = {"PHONE_DETECTED", "CHEAT_SHEET", "COMMUNICATION"}

async def run_instant_surety_check(
    exam_id: str,
    event_id: str,
    frame_path: str,
    candidate_category: str,
    cv_confidence: float,
    bbox: dict | None = None
):
    """Runs immediately and asynchronously after 1A candidate triggers."""
    prompt = build_instant_surety_prompt(candidate_category, cv_confidence, bbox)
    result = await vlm.execute_single_frame(image_path=frame_path, prompt=prompt)

    required_confidence = 0.92 if candidate_category in NON_DISMISSIBLE_CATEGORIES else 0.85

    if not result.get("plausible") and result.get("confidence", 0.0) >= required_confidence:
        # Confident false positive — fast clear immediately
        await update_event_status(event_id, AnomalyStatus.AI_CLEARED, source="SURETY_CHECK_FAST_CLEAR")
        event = await get_event_async(event_id)
        await broadcast_event(exam_id, event)
```

Background Auto-Enrichment Scheduler

```
async def schedule_tier3_enrichment(exam_id: str, event_id: str, trigger_timestamp: float):
    """Waits until 35s after trigger so the full 60s temporal window exists in buffer."""
    elapsed = time.time() - trigger_timestamp
    wait_needed = max(0.0, 35.0 - elapsed)
    await asyncio.sleep(wait_needed)

    event = await get_event_async(event_id)
    if event.status == AnomalyStatus.AI_CLEARED:
        # Fast-cleared by Surety Check — skip Tier 3 entirely, saving GPU compute
        return

    await run_tier3_auto_enrichment(exam_id, event_id)
```

5. Frontend UI/UX Specification

Flicker-Free Micro-Staging in AlertPanel.js

1. **Card Status Badges**:
   - `PRELIMINARY`: Amber pill + `<Clock className="animate-spin" /> Verifying...`
   - `AI_CONFIRMED`: Crimson pill + `<ShieldAlert /> Confirmed Violation`
   - `AI_CLEARED`: Green pill + `<CheckCircle2 /> Benign (Cleared)`
   - `NEEDS_REVIEW`: Purple pill + `<AlertTriangle /> Needs Proctor Review`
   - `HUMAN_RESOLVED`: Slate pill + `<UserCheck /> Resolved by {proctor_name}`
2. **Micro-Staging**: 1A triggers appear with a subtle staging animation. If fast-cleared within 1.5s, they slide silently into `Cleared Logs` without creating audio/visual alert fatigue.
3. **Deck Filtering**: `Active Infractions` vs. `Cleared Logs` (Fast-cleared + Tier 3 cleared) vs. `Resolved`.
4. **Action Bar**: **"Rerun AI Analysis"** (Tier 2) and **"Deep Analysis"** (Tier 3b), disabled with a lock icon while `is_reanalyzing === true`.

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6. Continuous Learning & Reinforcement from Human Feedback (RL / DPO)

While pre-trained VLM weights remain frozen during live inference, ARGUS implements a **Continuous Feedback Flywheel** to systematically learn from invigilator corrections:

```
┌──────────────────────────────────────────────────────────────────┐
│                        HOW ARGUS LEARNS OVER TIME                │
├──────────────────────────┬───────────────────────────────────────┤
│ Level 1: Immediate       │ In-Context "Mistake Memory" Prompt Injection│
│ Level 2: Periodic Batch  │ DPO / LoRA Adapter Fine-Tuning (~30 mins)   │
└──────────────────────────┴───────────────────────────────────────┘
```

Level 1: In-Context "Mistake Memory" (Immediate Dynamic Feedback)

1. **Tagging Corrections**: When an invigilator marks an event as `FALSE_POSITIVE` or edits a classification in the UI, ARGUS records a structured correction record into `human_feedback_dossier`:

```
{
  "exam_id": "exam_402",
  "camera_id": 1,
  "desk_id": 4,
  "cv_flag": "PHONE_DETECTED",
  "human_ground_truth": "FALSE_POSITIVE",
  "reason_tag": "PEN_MISIDENTIFICATION",
  "correction_notes": "Candidate holding a thick black cylindrical fountain pen."
}
```

2. **Dynamic Calibration Injection**: For subsequent frames in that room/hall, the backend queries recent human corrections and injects a dynamic calibration block into the VLM prompt:

```
HALL CALIBRATION (Verified Human Proctor Corrections in this Venue):
- Desk 4: Thick cylindrical drawing pens were previously misidentified as phones.
- Desk 12: Dark knitted winter headwear must be marked as 'obstructed' visibility, NOT 'earbud_detected'.
```

3. **Outcome**: The VLM immediately avoids repeating the exact same false positive without touching neural network weights.

Level 2: DPO & LoRA Fine-Tuning (Periodic Offline Model Updates)

Classical PPO (Reinforcement Learning with Human Feedback) requires training a separate reward model, which is unstable for vision models. Instead, ARGUS implements **Direct Preference Optimization (DPO)**, the modern standard for reinforcement learning in VLMs.

1. **DPO Pair Assembly**: Every proctor resolution creates a training pair:
   - **Context (x)**: Surveillance keyframe image.
   - **Rejected Response (y_l)**: Model's incorrect prediction (`phone_detected: true`).
   - **Chosen Response (y_w)**: Proctor's ground truth (`phone_detected: false, notes: holding pen`).
2. **Periodic LoRA Fine-Tuning**: Once 200–500 proctor feedback pairs accumulate:

- Run an automated LoRA fine-tuning script on Qwen3-VL: $\mathcal{L}_{DPO}(\theta) = -\mathbb{E}_{(x, y_w, y_l)} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(y_w|x)}{\pi_{ref}(y_w|x)} - \beta \log \frac{\pi_\theta(y_l|x)}{\pi_{ref}(y_l|x)} \right) \right]$
- Training takes **~30–45 minutes** on a single consumer GPU (RTX 3060/4070).

3. **Hot-Swapping**: Export the lightweight LoRA adapter file (`~50 MB`) and mount it to LM Studio or Ollama. The model permanently internalizes the venue's camera angles, lighting, and student attire.

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7. Enterprise Security, Concurrency & Audit Rules

Rule	Implementation Requirement
Data Immutability	Never overwrite original Tier 1 event records. Re-analyses append linked JSON records inside `analysis_data["history"]`.
Concurrency Locking	Set `is_reanalyzing = true` on the DB record. Disable action buttons for all connected proctors to prevent duplicate submissions.
Audit Trail	Record `user_id`, role, client IP, timestamp, and prompt tier for every action; record `source: SURETY_CHECK_FAST_CLEAR` / `system:tier3_auto` for automated ones.
Rate Limiting	Server-side enforcement: max 3 Tier 2 re-runs; max 1 Tier 3b Deep Analysis per event.
WebSocket Isolation	Broadcast anomaly status changes only to clients in the specific `exam_id` room, not globally.
Error Handling	Timeouts/parse errors trigger a distinct `ERROR` status with retry actions — never coerced into `NEEDS_REVIEW`.

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8. Hardware Latency, GPU Throttling & Resource Benchmarks

Operation	Input Size	Local GPU (Qwen3-VL 4B / RTX 3060/4070)	Cloud API (Gemini 2.0 Flash)
Tier 1A (CV Detection)	1 Frame (1080p)	**18 – 35 ms**	N/A (runs locally on edge)
Instant Surety Check	1 Frame, 4-field schema	**1.0 – 1.8 seconds**	**0.5 – 1.0 seconds**
Tier 1B (Adaptive VLM Sweep)	1 Frame (Round-Robin)	**1.5 – 2.2 seconds**	**0.8 – 1.5 seconds**
Tier 2 (Single-Frame Re-run)	1 Frame (Keyframe)	**1.5 – 2.5 seconds**	**0.8 – 1.5 seconds**
Tier 3 (60s Auto Video)	20–24 Sampled Frames	**4.5 – 7.5 seconds**	**2.5 – 4.0 seconds**
Tier 3b (Densified Deep)	35–40 Sampled Frames	**7.5 – 12.0 seconds**	**4.0 – 6.5 seconds**

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9. Engineered Mitigations Summary

1. **GPU & API Cost Saturation**: Resolved via Adaptive Round-Robin Scheduling for Tier 1B (~1 check/10–15s per desk) and concurrency limiting (`max_workers=2`).
2. **UI Card Flicker**: Resolved via 1.5s Micro-Staging — fast-cleared false positives slide into `Cleared Logs` without flashing the active alert deck.
3. **Memory Footprint (14GB → 350MB)**: Resolved by encoding frames to compressed JPEG byte strings inside the 75-second rolling buffer.
4. **False-Clear Cheating Vulnerability**: Resolved by enforcing a strict confidence threshold (`>= 0.92`) and disallowing fast-clears on concealed item categories (`PHONE_DETECTED`, `CHEAT_SHEET`, `COMMUNICATION`).
5. **Continuous Learning from Mistakes**: Resolved via dynamic in-context prompt calibration (immediate) and periodic DPO LoRA adapter updates (monthly).